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Toward the Development of an NH_3-RRQL System—Part II: Effects of the Primary Combustion Zone Length and Secondary Stage Number of Holes on Stability and Emissions

Staged combustion systems like the rich–relaxation–quench–lean (RRQL) offer the potential for low nitrogen oxides (NO_x) emissions while burning ammonia (NH_3). This process involves rich premixed NH_3–air combustion in a primary zone, allowing time for NO_x relaxation, followed by a lean secondary combustion zone via air injection. However, improper design of the secondary stage can lead to high NO_x and nitrous oxide (N_2O) emissions, offsetting the climate benefits of carbon-free fuels. This study investigates the effects of primary zone length and secondary stage geometry on the stability and emissions of a lab-scale RRQL system. Experiments were conducted at atmospheric pressure using a modular axial swirl burner (swirl number = 1.1, 16 vanes), with primary equivalence ratios ($\phi_{primary}$) of 1.13, 1.15, and 1.18. Two quartz lengths (76 and 178 mm) were tested using a five holes (2.03 mm) secondary injection design. Strong flame interaction and elevated NO_x were observed for the 76 mm liner. A longer chamber allowed better NO_x relaxation and NH_3 cracking. Additional tests with 5-, 10-, and 16-holes configurations showed that fewer holes, implying higher momentum flux ratios, yielded lower NO_x–N_2O emissions, especially at $\phi_{primary} = 1.13$. These effects diminished as $\phi_{primary}$ decreased from 1.18 down to the optimum 1.13. Diffusion-like combustion was seen for $0.90 \leq \phi_{global} \leq 1.10$, leading to inefficient combustion marked by excess O_2 values compared with equilibrium. Optimal performance was achieved with $\phi_{primary} = 1.13$ and $0.70 \leq \phi_{global} \leq 0.90$, with estimated burner outlet temperatures between 1720 and 1970 K. [DOI: 10.1115/1.4069538]

Introduction

Hydrogen (H_2) as fuel for electrical power generation is an important vector toward a zero-carbon economy, if solutions are found to alleviate its main limitations, including low volumetric density, high risk of explosion, and storage and transport [1]. Instead, ammonia (NH_3), a H_2 carrier, has attracted a lot of attention in recent years owing to its mature and well-established infrastructures for production, storage, and distribution [2,3]. Many countries, such as Japan, South Korea, Australia, and U.S., to name a few, are now funding research programs to promote NH_3 for energy. The major challenges of burning NH_3 are its very low reactivity, which can be overcome by blending with methane (CH_4) or H_2 to stabilize the flame [4–6], and the very high NO_x emission [7–9]. For example, researchers found that the NO_x emission level is unacceptably high when burning NH_3 at the conditions under which most gas turbines operate [10–13]. Moreover, NH_3 is toxic and corrosive, raising additional difficulties when operating combustion devices with this fuel. However, blending NH_3 with other fuels could be challenging from an infrastructural point of view, and very few studies are available about pure NH_3–air combustion for gas turbines.

Rich NH_3–air and NH_3–H_2/CH_4–air premixed flames with equivalence ratios around 1.10 to 1.30 are known to yield marginal NO_x emissions without needing to resort to selective catalytic

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reduction [13,14]. However, these conditions lead to unburned NH_3, H_2, and/or CH_4, so a second combustion stage is needed to oxidize all the fuel. This strategy is typically referred to as rich–quench–lean [15–17] or two-stage combustion [18,19], which relies in a NH_3 flame with rich primary equivalence ratio ($\phi_{primary}$) to produce low NO_x, and then a secondary air is injected to burn out the remaining fuel in a secondary lean combustion zone, producing a lean global equivalence ratio (ϕ_{global}).

Tu et al. [20] experimentally and numerically studied NH_3–CH_4–air premixed swirling flames with air-staging showing that using 30% of the air budget for the secondary stage, which produced a $\phi_{primary} = 1.21$, was the optimum staged-air ratio for low NO emissions and high combustion efficiency. However, they demonstrated that NO emissions could reach high values either by injecting secondary air too close to the primary flame or by setting too high staged-air ratios. Finally, high jet-to-crossflow momentum flux ratio (J) could produce air jets impinging on each other, creating an upstream recirculation which was found to negatively affect the NO emissions. Ånestad et al. [21] investigated the flame structure and emissions performance in a two-stage lab-scale generic combustor design operated with NH_3–H_2 blends. They found minimum NO_x emissions at $\phi_{primary} \approx 1.15$, which occurred when excess NH_3 in the main stage was largely converted through pyrolysis to H_2, which was then burnt off with additional air in the second stage. However, in both cases, using constant fuel and air budgets and then splitting the air into primary and secondary streams changes the $\phi_{primary}$ for each splitting ratio, making it difficult to isolate the effects of diverse parameters to compare between cases, and conclude about the sole impact of the secondary stage on exhaust emissions and flame stability.

Avila Jimenez et al. [22] investigated the effects of secondary air injection on the emissions and stability of two-stage NH_3–CH_4–air swirl flames using a lab-scale microgas turbine burner. They demonstrated that having more secondary air holes for the same air addition could reduce NO emissions. In addition, they demonstrated for the first time that there is a secondary air addition threshold (which depends on the secondary air injection geometry) above which primary flame instabilities are triggered, which increased the NO emissions prior to blowout. Recently, Kim et al. [23] explored how secondary air injection impacts the flame structure and NO emission characteristics in turbulent premixed CH_4–NH_3–air flames within a two-stage swirl model combustor. They discovered that minimal NO emissions, along with low CO emissions, occur when there is weak interaction between the two combustion zones and when the secondary air jet has a moderate velocity. Nonetheless, there has been limited investigation into the effects of other parameters on flame stability beyond just the secondary air flow rate.

Takagi et al. [24] performed staged combustion experiments by adding NH_3 to a hydrocarbon (C_3H_8) and nonhydrocarbon ($CO + H_2$) fuel blend. They demonstrated that the conversion of NH_3 (only product for NH_3–nonhydrocarbon blend) into NO in the lean secondary combustion stage is much lower than that for HCN (additional product from NH_3–hydrocarbon blend). Therefore, mixing hydrocarbons to stabilize rich NH_3 flames could not be a good strategy due to the possible HCN emissions from the rich primary stage that can potentially increase the NO formation in the secondary stage.

Cole et al. [25] recently introduced the rich, relaxation, quench, lean (RRQL) strategy for NH_3 combustion. The experimental setup designed in Ref. [25] was upgraded with a redesigned modular swirler, for which the procedure is discussed in Part I of this study. This paper (Part II) presents an experimental study of the impact of the combustion chamber length and the secondary air injection number of holes on combustion stability and exhaust emissions. First, two quartz liners with 76 mm and 178 mm length were implemented for the primary combustion zone while injecting secondary air through a secondary stage geometry with five holes of 2.03 mm diameter. Then, three different secondary stage geometries with 5-, 10-, and 16-holes were tested with the 178 mm long quartz were studied.

Materials and Methods

Burner and Conditions. Experiments were conducted with a laboratory scale modular burner, shown in Fig. 1. The burner consists of a modular swirler, and the swirler center body houses a hydrogen torch and spark ignitor which is used only to ignite the premixed NH_3–air. This burner simulates the rich, relaxation, and lean secondary stage combustion sections of an RRQL NH_3 combustor, and the reader can refer to Ref. [25] for further details about the burner head end. The burner has been designed also to

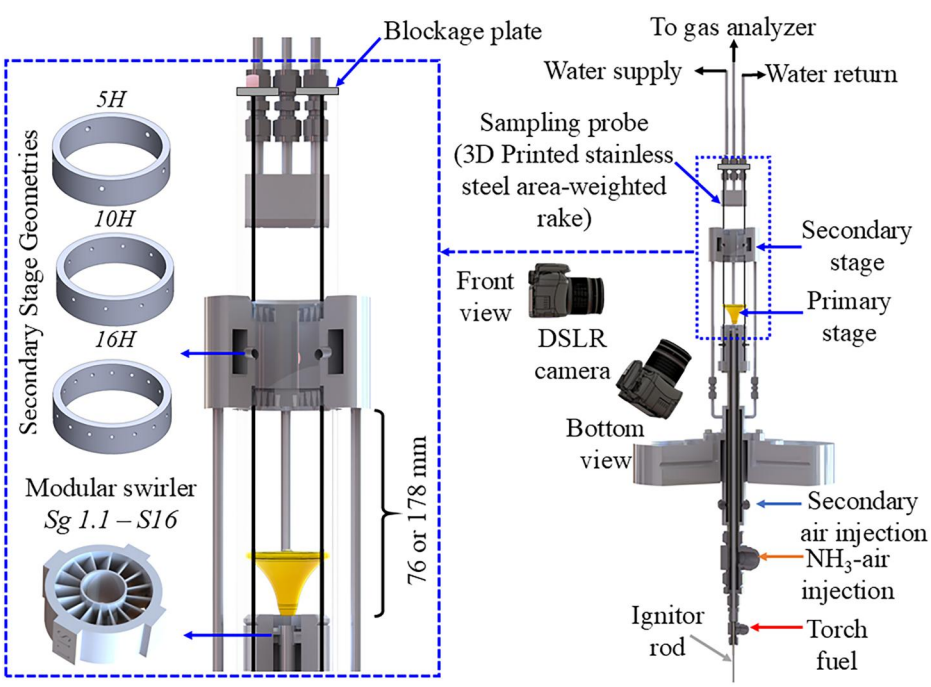

Fig. 1 Modular swirl burner setup

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accommodate a 3D printed modular and interchangeable swirler manufactured in stainless steel. In this study, a swirler design with 16 straight vanes with a geometrical swirl number of $S_g = 1.1$ was chosen based on previous experiments (see Part I of this paper). The swirler has 16 straight vanes with a vane angle of 55.95 deg and an effective area of 6.7×10^{-4} m^2 and is illustrated in Fig. 1.

Two experimental campaigns were conducted. First, the combustion chamber length impact on emissions and flame stability was studied. Two quartz lengths of 76 mm and 178 mm were implemented for the primary combustion zone while injecting secondary air through a secondary stage geometry of five holes with 2.03-mm diameter, as shown in Fig. 1. The ϕ_{global} was decreased to around 0.50 or flame blowout. Time-averaged broadband flame images of the primary combustion zone were recorded with a DSLR camera to characterize the interaction between the secondary and primary flames. The camera was set with an exposure time of 1/8 s, an aperture of *f*/8, and an ISO number of 800. White balance was kept constant at 4000 K (white, fluorescent light) throughout the experimental campaigns.

Second, the exhaust emissions for different number of holes in the secondary stage were investigated. Three different secondary stage geometries with 5-, 10-, and 16-holes were tested with the 178 mm long quartz. Time-averaged broadband flame images of the secondary combustion zone were recorded with the same DSLR camera from a bottom view (see Fig. 1, right-hand side) to characterize the secondary combustion zone. The camera settings were changed to an exposure time of 2 s, an aperture of *f*/8, and an ISO number of 1600.

Equivalence ratios and the nozzle velocity are calculated in real-time using the mass flow ratios for the fuel and air circuits. The mass flow rates are metered by pressure drop through critical orifices and controlled using manual and electronic pressure regulators to control upstream pressure. An upstream pressure transducer and thermocouple in combination with a downstream pressure transducer allow the operator to ensure sonic flow for effective metering. An additional thermocouple is located upstream of the swirler to track the reactant temperature. Constant nozzle velocity of $U_0 = 0.90$ m/s was utilized and defined using the reactant density, effective nozzle area, and reactant mass flow. The current air and fuel flow control setup leads to accuracy for the ϕ_{global} better than 1%.

Table 1 summarizes the experimental procedure followed. There is a blockage plate on the top of the top quartz (see Fig. 1) to prevent backflow or interactions with surrounding air. The long quartz is meant to allow NO_x emissions to relax toward equilibrium with residence times around 100 ms, depending on the nozzle velocity and expansion ratio. As shown in Ref. [26], these values should enable the capture of the first order of magnitude NO relaxation at 1 bar and nearly the full approach to equilibrium at 10 bar.

Exhaust Emissions. Exhaust emissions data are collected using a California Analytical Instruments 700 LX Series NDIR/O_2 Analyzer for O_2 and 700 FTIR multigas analyzer with NO_x, NH_3, and N_2O measurement capabilities. An area-weighted, water-cooled rake probe was installed 76 mm downstream from the secondary stage holes plane, and the reader is referred to Ref. [25] for additional information about the sampling probe design. The probe is complete with thermocouples on the water supply, water return, and sample lines to ensure that the sample is properly quenched to freeze NO_x reaction chemistry and to avoid water condensation in the sampling line. Additionally, the sample temperature is closely monitored, and the emissions sample line between the probe and analyzer is heated to 150 °C. Exhaust emissions were normalized by the standard of 15% dry O_2 [27]. All the raw (non-normalized by the standard of 15% dry O_2) emissions data are plotted as Supplemental Materials on the ASME Digital Collection. The exhaust emission gas analyzers are calibrated before every run using span gas with a known concentration for O_2, CO, CO_2, NO, NO_2, N_2O, and NH_3. In addition, exhaust emissions concentration is contrasted using a standard well-known flame every time before starting an experimental campaign. By this process, it is confirmed that the accuracy of the reported data is within the accuracy of the gas analyzers provided by the company (California Analytical Instruments, Orange, CA). Unfortunately, the combustion efficiency could not be calculated due to lack of H_2 capability in the current gas analyzers setup. As an alternative for this metric, the equilibrium O_2 was computed and compared with the measured values. In this sense, higher combustion efficiency is rated by the proximity of measured O_2 to equilibrium O_2, i.e., higher measured O_2 than equilibrium indicates lower combustion efficiency.

Results and Discussion

Long Versus Short Quartz. Figure 2 shows the concentration of NO_x (top panel), N_2O (middle panel), and NH_3 (bottom panel) in the exhaust as a function of ϕ_{global} for the two combustion liners' length tested (76 mm and 178 mm). Solid and dashed lines show the results for long and short liners, respectively. Lower operability range was obtained with the short liner compared to the longer one. The lowest ϕ_{global} achieved with the short liner before primary flame blowout was $\phi_{\text{global}} = 0.76 \pm 0.02$, regardless of the ϕ_{primary} tested, although exhaust emissions were recorded only until $\phi_{\text{global}} = 0.80$ (or higher) due to primary flame instabilities. These flame instabilities occurred due to strong interaction between the secondary stage with the primary flame (see Fig. 3). In contrast, stable primary flame was observed until $\phi_{\text{global}} = 0.50$ with the long liner ($\phi_{\text{global}} < 0.50$ could not be studied due to air supply limitations). This means that longer combustion liner is desirable for RRQL NH_3 combustion system since the lean stability limit is highly increased. It is also concluded that implementing a short liner, which is closer to a retrofitted burner, not only carries flame instabilities for lean ϕ_{global} but also higher NO_x emissions.

Figure 2 shows that very low NO_x emissions were measured for the rich flames (no secondary air addition) regardless of the combustion liner length. However, higher NO_x concentration was measured with the short liner compared to the long one at the same ϕ_{global} when secondary air was supplied. One explanation, as described later by analyzing primary flame images, is that strong interactions between the secondary and primary combustion zones may produce local lean pockets, conditions that are well-known to

Table 1 Experimental matrix

Parameter	Campaign 1	Campaign 2
Primary equivalence ratio (ϕ_{primary})	1.13/1.15/1.18	
Nozzle velocity (U_0)	0.90 m/s	
Geometric swirl (S_g)	1.1	
Inlet temperature	298 K	
Inlet pressure	0.11 MPa	
Ammonia mol fraction (X_{NH3})	1	
Global equivalence ratio (ϕ_{global})	1.10 down to 0.50 or blowout	1.10 down to 0.50
Quartz length (mm)	76 and 178	178
Secondary stage # holes	5	5/10/16
Secondary stage holes diameter (mm)	2.03	

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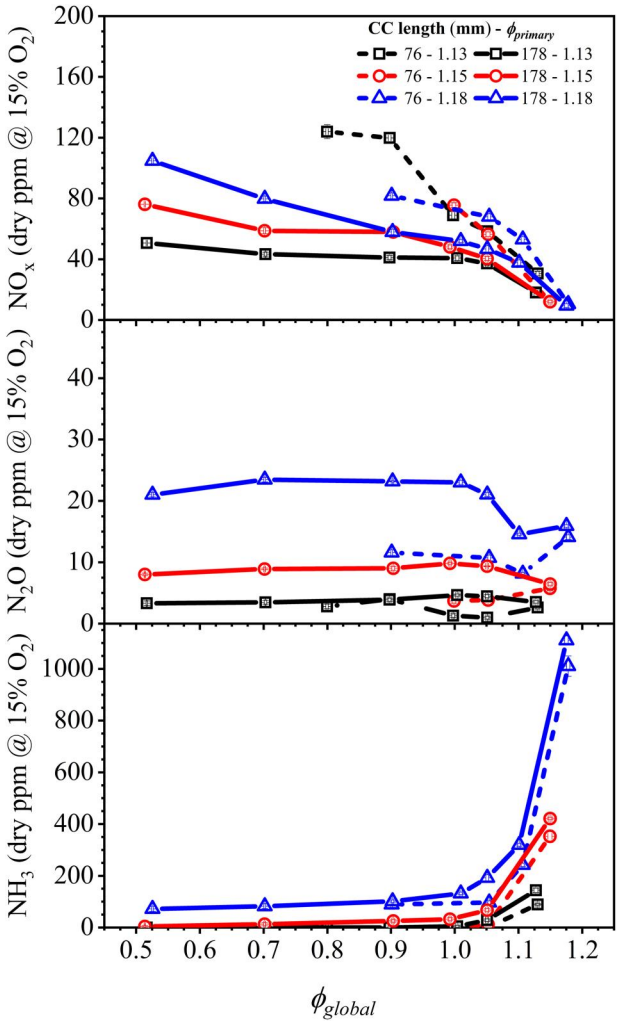

Fig. 2 NO_x (top panel), N_2O (middle panel), and NH_3 (bottom panel) concentration in the flue gases as a function of the global equivalence ratio for 76 mm (dashed lines) and 178 mm (solid lines) quartz length. Primary equivalence ratio is shown in different colors and symbols ($\phi_{primary}$ = 1.13 squares, $\phi_{primary}$ = 1.15 circles, and $\phi_{primary}$ = 1.18 triangles).

produce high NO_x emissions [7–9,13,28,29]. Kim et al. [23] explain that local lean pockets could be produced, increasing the NO emissions, when strong interaction happens between the air jets in the secondary combustion zone and the primary rich flame, and as shown in Ref. [20], the NO emissions increased by placing the secondary air addition too close to the primary flame. On the other hand, lower N_2O is measured for the short liner (Fig. 2, middle panel). This is probably due to higher temperatures and lower heat losses [30] compared to the long liner, a trend that is also observed for NH_3 emissions. This also implies that probably a slightly lean flame is produced with the short liner when adding secondary air, explaining again the higher measured NO_x concentration. Although slightly higher N_2O and NH_3 emissions are produced with the long liner, it is observed that N_2O emissions are sub-10 ppm and NH_3 are negligible for $\phi_{primary} = 1.13$ and 1.15 for $\phi_{global} < 1.0$. Therefore, the long liner is better not only for lower NO_x but higher flame stability. Finally, it is important to note that $\phi_{primary} = 1.13$ yields the lowest exhaust emissions with the long liner for all the lean ϕ_{global} tested.

Figure 3 presents the flame images of the primary flame for the short (top row) and long (bottom row) liners as a function of the ϕ_{global} for a $\phi_{primary} = 1.18$. This $\phi_{primary}$ was selected since it allows for easier secondary flame observation by the eye, but the same trend is followed by flames with $\phi_{primary} = 1.13$ and 1.15. It is observed that the secondary combustion zone, denoted by the curved white line, grows and travels upstream by decreasing ϕ_{global}, regardless of the combustion liner length. However, due to the reduced distance between both stages, the secondary combustion zone interacts with the primary flame while using the short liner. This interaction triggers pulsating flame instabilities that eventually lead to flame blowout. This behavior was reported recently by Avila Jimenez et al. [22] for a different combustor architecture. These interactions also could produce slightly lean pockets that might explain the higher NO_x [20,23] and lower NH_3 emissions measured (see Fig. 2), and produce higher local temperatures, explaining also the lower N_2O emissions. In contrast, the secondary combustion zone does not interact with the primary flame for the long liner as observed in the bottom row for a wider ϕ_{global} range, until 0.70. Interactions between both combustion zones were only noticeable at far lean conditions $\phi_{global} < 0.70$. Although flame instabilities were not observed, these interactions could still create local lean pockets, explaining the increase in NO_x at the leaner ϕ_{global} (see Fig. 2, top panel). Therefore, the next section (Number of Holes) is presented only for the 178 mm long quartz.

Number of Holes. Figure 4 shows the concentration of NO_x (top panel), N_2O (middle panel), and NH_3 (bottom panel) in the exhaust as a function of ϕ_{global} for the three geometries and $\phi_{primary}$ tested. The different colors are for different $\phi_{primary}$, and the solid, dashed, and dotted lines are for 5-, 10-, and 16-holes in the secondary stage, respectively. Once again, very low NO_x emissions (sub-12 ppm) were measured for the rich flames (no secondary air addition) regardless of secondary geometry. However, NO_x emissions increase by decreasing ϕ_{global} for all geometries. Even though there are no flame instabilities or strong combustion zones interactions, it is observed that higher NO_x in the secondary combustion zone is produced as a function of the $\phi_{primary}$. This is explained by the much higher unburned NH_3 (see Fig. 4, bottom panel) available for the secondary stage combustion zone as $\phi_{primary}$ increases. It is important to note that the improvement in NO_x emissions by implementing a longer combustor liner for the primary flame could be canceled by poor control of the $\phi_{primary}$ and poor design of the secondary combustion zone.

For most cases, a monotonic trend is observed for NO_x and N_2O (Fig. 4, middle panel). It was found that $\phi_{primary} = 1.13$ produced the lowest NO_x emissions for over the lean cases. In addition, the secondary stage geometry plays a role. Lower number of holes yields better NO–N_2O emissions. However, these differences in NO and N_2O emissions between geometries are more noticeable by increasing $\phi_{primary}$. This means that there is a lower impact of the secondary stage geometry on emissions when $\phi_{primary}$ is tuned.

Figure 4 (bottom panel) shows that the unburned NH_3 is rapidly consumed in the secondary combustion zone for $\phi_{primary} = 1.13$ and 1.15, conditions that yield zero NH_3 emissions. However, unburned NH_3 is still measurable for $\phi_{primary} = 1.18$, explaining the higher NO and N_2O emissions for this condition. It is acknowledged that high heat losses through the walls can be responsible for part of the NH_3 emissions and lowering the heat losses by implementing a metal liner and using high preheat temperatures could bring NH_3 even lower and enhance the H_2 production for the secondary stage. It is concluded that $\phi_{primary} = 1.13$ is the optimum primary equivalence ratio, and $0.70 \leq \phi_{global} \leq 0.90$ seems to be an optimum operability range for the RRQL system using the current burner. This ϕ_{global} range produces a theoretical burner outlet gas temperature of around $1720\,K \leq \phi_{global} \leq 1970\,K$. It is acknowledged that gas turbines, especially the ones fitted with dry low NO_x combustors, run at lower ϕ_{global} to control turbine inlet temperature (TIT). However, TIT could be modified in practice by placing a dilution stage downstream the secondary lean combustion stage, which is out of the scope of the present study. Nevertheless, the authors acknowledge that with NH_3 as fuel, the lower ϕ_{global} studied are relevant to J-class gas turbines.

Figure 5 presents the time-averaged images of the secondary combustion zone for the three geometries studied as a function of the ϕ_{global} for a $\phi_{primary} = 1.13$. Flame images for $\phi_{primary} = 1.15$ and

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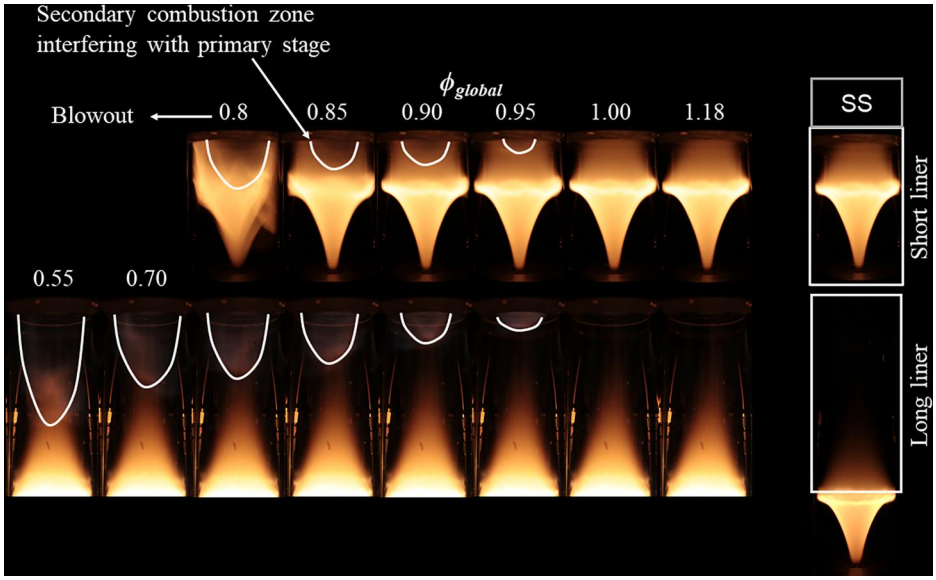

Fig. 3 Time-averaged broadband images of NH_3–air flames as a function of the global equivalence ratio for short (top row) and long (bottom row) combustion chamber

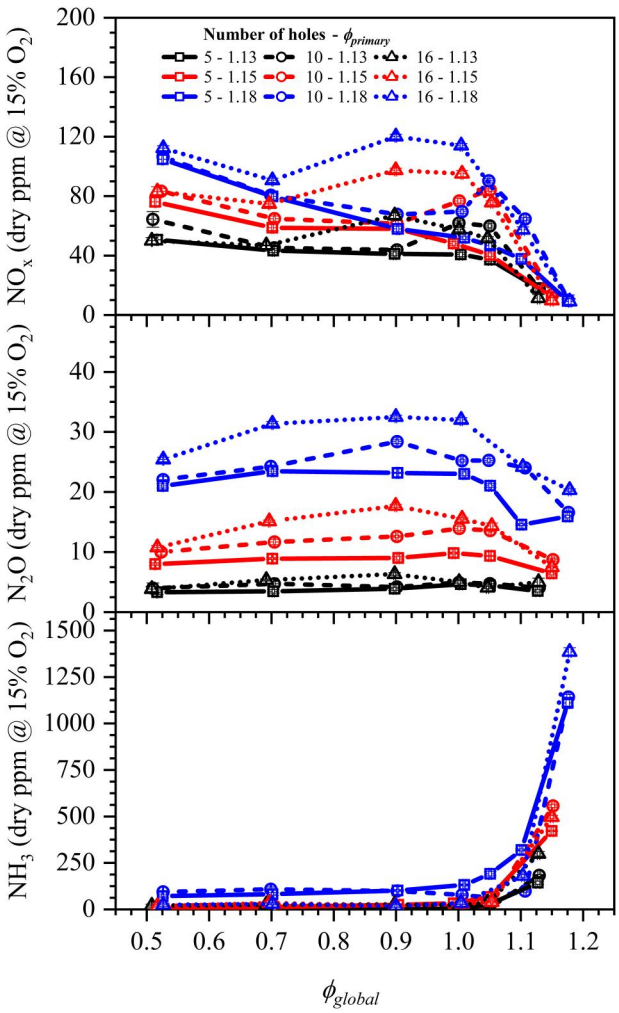

Fig. 4 NO_x (top panel), N_2O (middle panel), and NH_3 (bottom panel) concentration in the flue gases as a function of the global equivalence ratio for 5 (square, solid lines), 10 (circles, dashed lines), and 16 (triangles, dotted lines) holes in the secondary stage. Primary equivalence ratio is shown in different colors. (Color version online.)

1.18 are provided as Supplemental Materials on the ASME Digital Collection. Inverse diffusion flames are first formed, as highlighted by the blue rectangles (first rectangle from right to left). By comparing the NO_x (top panel) and NH_3 (bottom panel) emissions in Fig. 4 with the flame images in Fig. 5, it is observed that this combustion regime (diffusion-like) in the secondary stage is undesirable for either low combustion efficiency or high NO_x production. It is also notice that these "diffusion-like" flames are stabilized for leaner ϕ_{global} by increasing the number of holes, i.e., low J values are not appropriate.

Then, there is a transition to "partially premixed-like" combustion regime as shown by the arrow (see the yellow rectangles, second rectangle from right to left), which is characterized by a more homogeneous combustion zone, which ϕ_{global} depends on the secondary geometry. This transition occurs by decreasing ϕ_{global}, i.e., by increasing the secondary air flowrate, hence, increasing the jet's penetration length. This explains why there is an "optimum" ϕ_{global} and why it shifts to leaner values by increasing the number of holes. Contrasting again Figs. 4 and 5, it can be concluded that this combustion regime is desirable due to a local minimum NO_x and lower NH_3 emissions. Then, decreasing further ϕ_{global}, i.e., increasing J, produces a relocation of the secondary combustion zone upstream until it interacts with the primary flame (see white rectangle, far left rectangle) as it was explained in Fig. 3, increasing NO_x emissions again.

The comparison of flame images and exhaust emissions indicates that partially premixed-like combustion, with a ϕ_{global} slightly leaner than the transition value from diffusion-like (avoiding the interaction between both combustion zones), is an optimal secondary combustion process. It is also concluded based on the current results that high J that allows for a fast transition from diffusion-like to partially premixed-like combustion regime is key for the design of the secondary air injection geometry for an efficient RRQL system.

Figure 6 shows the concentration of CO (top panel) and O_2 (bottom panel) in the exhaust as a function of ϕ_{global} for the three geometries and ϕ_{primary} tested. CO emissions have been demonstrated for rich NH_3–air flames [25], due to the dissociation of the CO_2 in the atmospheric air, which is about 380 ppm [31]. CO reaches a concentration of about 40 ppm in the flue gases; however, it is quickly burned out in the secondary stage to sub-10 ppm values. As explained in the methods section, equilibrium O_2 is computed as a "combustion efficiency" measure. As can be seen in Fig. 5, diffusion-like combustion was observed for diverse $\phi_{\text{global}} \geq 0.85$,

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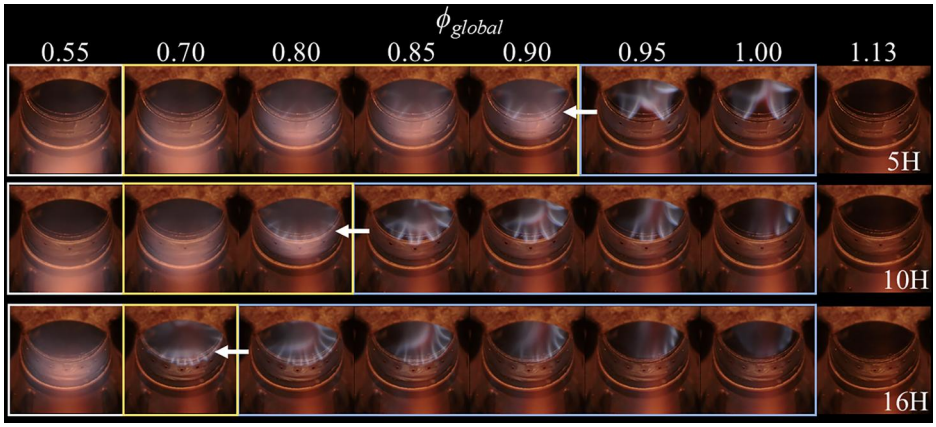

Fig. 5 Time-averaged broadband images of the secondary combustion zone as a function of the global equivalence ratio for the three geometries tested

depending on the number of holes. This combustion regime produces low combustion efficiency denoted by the high O_2 concentration compared with equilibrium O_2 in Fig. 6. All geometries follow the same trend and reach equilibrium O_2 at $\phi_{\text{global}} \leq 0.70$. For these lean ϕ_{global} values, a "premixed-like" combustion zone is observed (see Figs. 3 and 5), which stabilizes around 25 mm upstream the secondary holes plane.

Finally, an interesting comment can be made if NO_x emissions are plotted as a function of J as shown in Fig. 7. Two distinct trends are observed depending on the hole's setup for the secondary combustion zone. NO_x emissions keep increasing as a function of J for 5H, while there is a bimodal trend for 10H and 16H, with a local minimum NO_x concentration at certain J. This observation indicates that an odd number of holes in the secondary zone, meaning air jets do not face each other, would yield different NO_x trends compared to even number of holes (opposed air jets). In addition, Fig. 7 shows that the local minima for 10H and 16H is measured for $20 < J < 100$. Lower J leads to low combustion efficiency (see Fig. 6, bottom panel) and increased NO_x emissions, while higher J could also lead to increased NO_x emissions. It is suggested that the local minima happens when the hot products and secondary air are well mixed, and then the emissions increase again when the backflow starts interacting with the primary stage, producing local lean pockets [23], conditions that are well-known for producing high NO_x emissions for NH_3 flames [7–9,13,28,29]. However, these results could help as a design criterion to choose the number of holes for the secondary combustion zone in NH_3-RRQL systems.

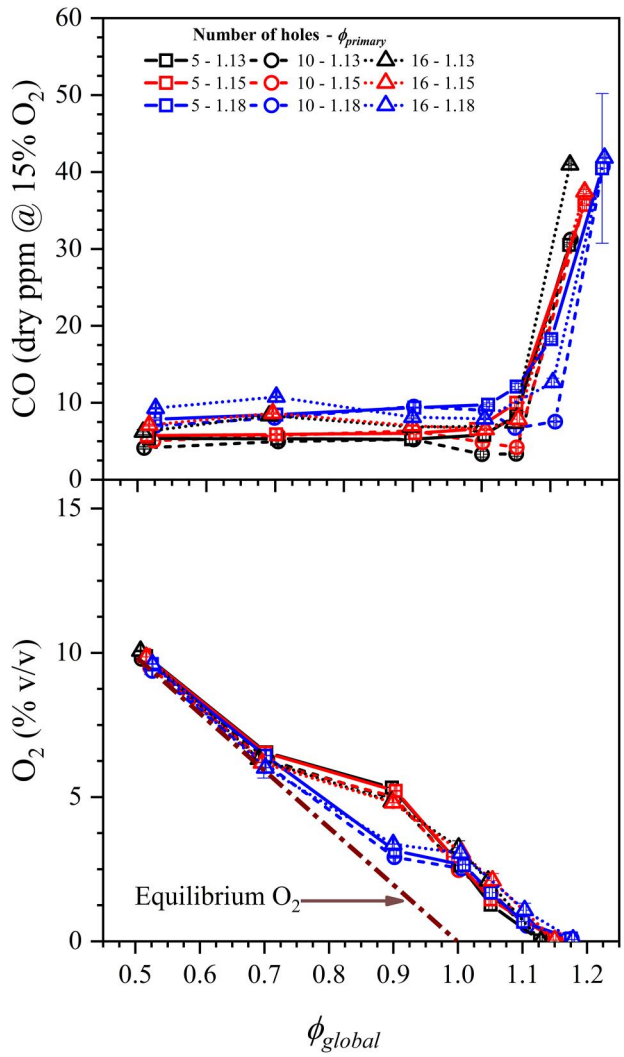

Fig. 6 CO and O_2 concentration in the flue gases as a function of the global equivalence ratio for different number of holes in the secondary stage. Primary equivalence ratio is shown in different colors. (Color version online.)

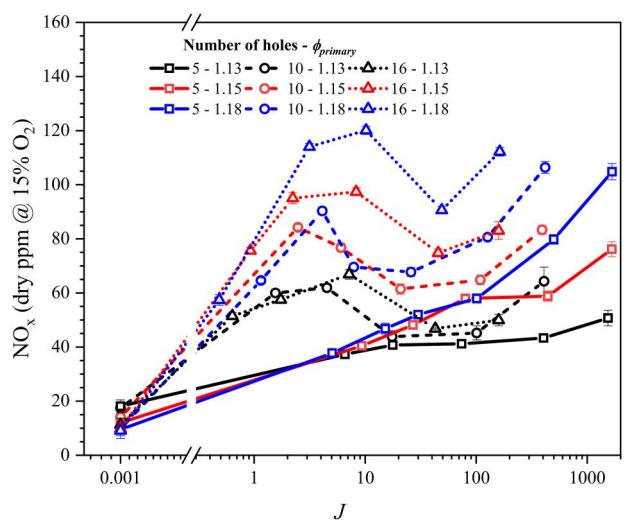

Fig. 7 NO_x concentration in the flue gases as a function of the momentum flux ratio (J) for different number of holes in the secondary stage. Primary equivalence ratio is shown in different colors. (Color version online.)

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Conclusions

An experimental study was carried out to understand the impact of the primary combustion zone length and secondary stage number of holes on stability and emissions of rich premixed NH_3–air flames under the RRQL system. The main conclusions are as follows:

- Long combustion chamber for the primary flame is better for lower NO_x and higher primary flame stability. This reduces the primary and secondary flames interaction which produces first higher NO_x emissions and then primary flame blowout if ϕ_{global} is reduced from a certain threshold (approximately $\phi_{global} = 0.76$ for a 76 mm combustor liner).
- For the tested burner, an optimum primary equivalence ratio was found to be $\phi_{primary} = 1.13$. This value not only allowed for a stable primary flame but also produced simultaneous low NO_x and NH_3 emissions.
- The secondary stage geometry plays a role on overall exhaust emissions. It was found that lower number of holes yields better NO_x–N_2O emissions. However, these differences between geometries observed for the exhaust emissions are more noticeable when increasing $\phi_{primary}$, probably due to the higher unburned NH_3 available for the secondary combustion zone.
- Diffusion-like combustion was found at low J values ($\phi_{global} \geq 0.85$, depending on the number of holes) producing higher O_2 concentration in the flue gases than expected from equilibrium calculations. This implies that such a combustion regime is not desirable due to lower combustion efficiency. Instead, premixed-like combustion was observed for higher secondary air additions, reaching an optimum operating range of $0.70 \leq \phi_{global} \leq 0.90$, which allows for a theoretical turbine inlet temperature range of TIT = 1720–1970 K.

The key takeaways are that a longer combustion chamber will not only increase the residence time for NO_x to reach equilibrium and crack unburned NH_3 but also increase the primary flame stability, increasing the operability range. In addition, we found that it is crucial to tune the rich primary flame equivalence ratio and then inject enough secondary air to avoid inverse-diffusion flames forming and instead reach a premixed-like secondary combustion zone for lower NO_x emissions.

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Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Nomenclature

CO = carbon monoxide
H_2 = hydrogen
J = jet-to-crossflow momentum flux ratio
NO = nitrogen oxide
NH_3 = ammonia
NO_x = nitrogen oxides
N_2O = nitrous oxide
O_2 = oxygen
RRQL = rich–relaxation–quench–lean
S_g = geometric swirl number
U_0 = nozzle velocity
ϕ_{global} = global equivalence ratio
$\phi_{primary}$ = primary equivalence ratio

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